

# Beryllium-9

<sup>9</sup>Be (Be I), nuclear spin  $I = 3/2$ . Energy levels and transitions below 2  $\mu\text{m}$ : 15 levels, 24 lines.

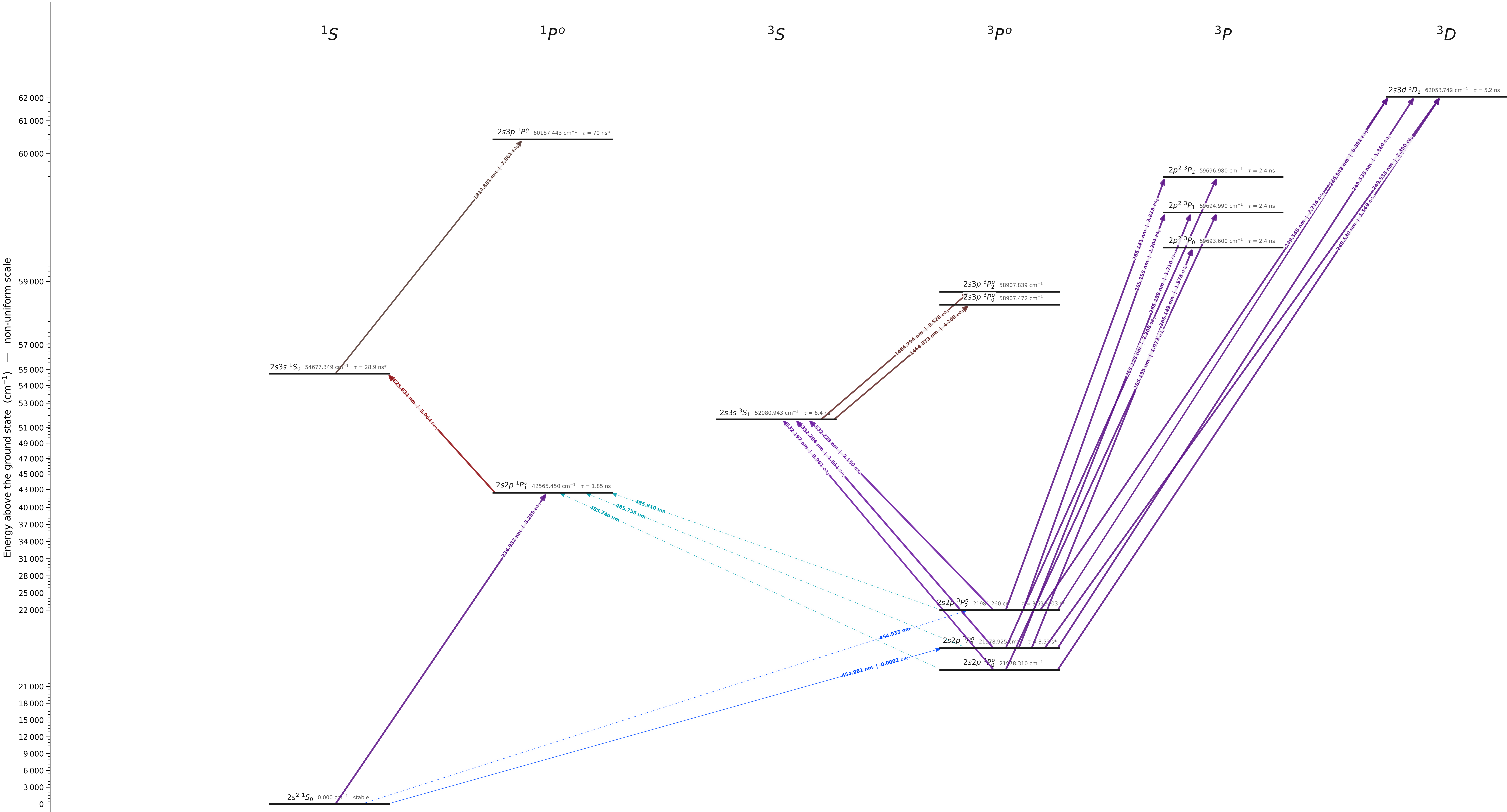
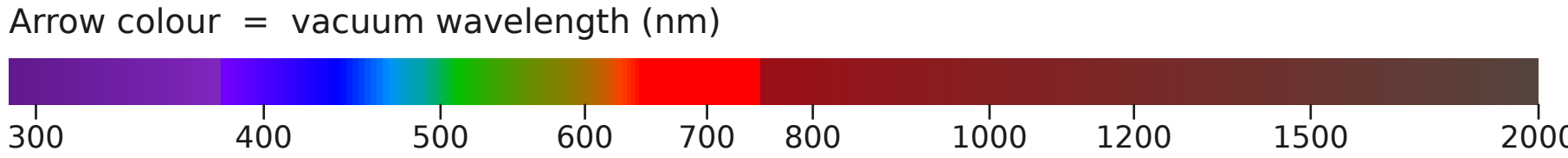
Be<sup>+</sup> ionisation limit 75192.64 cm<sup>-1</sup> = 9.32270 eV (132.992 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|\langle J \| er \| J' \rangle|$  in  $ea_0$

Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model calculation.

Arrow width grows with the Einstein coefficient  $A$ . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .

Level caption: NIST energy and measured lifetime (where one exists).



## Key transitions

| Transition                    | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz)   | $ U e U' $ (eao) | $A$ ( $s^{-1}$ ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use                            |
|-------------------------------|--------------------|--------------------|---------------|------------------|------------------|-------------------------|------------|--------------------------------|
| $2s^2\ ^1S_0 - 2s2p\ ^1P_1^o$ | 234.9323           | 234.8604           | 1276.08008(5) | 3.255            | 5.520e+08        | 86.03                   | 100        | 234.9 nm resonance line        |
| $2s^2\ ^1S_0 - 2s2p\ ^3P_1^o$ | 454.9813           | 454.8538           | 658.91016     | 0.000            | 4.200e+01        | 4.43e-08                | 100        | 454.9 nm intercombination line |
| $2s2p\ ^1P_1^o - 2s3s\ ^1S_0$ | 825.6342           | 825.4073           | 363.10565(13) | 3.064            | 3.380e+07        | 5.517                   | 97.5       | 825.6 nm                       |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available.  $A$ : measured value or NIST.  $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

### <sup>9</sup>Be hyperfine structure ( $I = 3/2$ )

| Level           | A (MHz)       | B (MHz)  | Shift of each F level from the centre of gravity (MHz) |                |                |                |
|-----------------|---------------|----------|--------------------------------------------------------|----------------|----------------|----------------|
| 2s2p $^3P_1^o$  | -139.373(12)  | -0.75(4) | F=1/2: +347.49                                         | F=3/2: +140.13 | F=5/2: -209.25 |                |
| 2s2p $^3P_2^o$  | -124.5368(17) | 1.429(8) | F=1/2: +561.67                                         | F=3/2: +373.61 | F=5/2: +61.38  | F=7/2: -373.25 |
| 2s2p $^1P_1^o$  | -13.836       | 0.845    | F=1/2: +35.65                                          | F=3/2: +12.99  | F=5/2: -20.54  |                |
| 2s3s $^3S_1$    | -332.250      | -        | F=1/2: +830.62                                         | F=3/2: +332.25 | F=5/2: -498.38 |                |
| 2p $^2$ $^3P_1$ | 18.101        | 0.734    | F=1/2: -44.34                                          | F=3/2: -18.84  | F=5/2: +27.34  |                |
| 2p $^2$ $^3P_2$ | 3.386         | -1.468   | F=1/2: -16.52                                          | F=3/2: -10.16  | F=5/2: -0.78   | F=7/2: +9.79   |
| 2s3p $^1P_1^o$  | -6.704        | 0.427    | F=1/2: +17.29                                          | F=3/2: +6.28   | F=5/2: -9.95   |                |
| 2s3d $^3D_1$    | 150.600       | 0.016    | F=1/2: -376.48                                         | F=3/2: -150.62 | F=5/2: +225.90 |                |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                    | $\lambda$ vac (nm) | Isotopes | Shift (MHz) |
|-------------------------------|--------------------|----------|-------------|
| $2s^2\ ^1S_0 - 2s2p\ ^1P_1^o$ | 234.932            | 11-9     | 16015.29(3) |

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$